

Surprisal Outperforms Other Automatically Computable Predictive Metrics

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Introduction

Hale (2001); Levy (2008)

$$\text{effort}(w_n) \propto \text{surprisal}(w_n) = -\log_2 P(w_n \mid w_1 \dots w_{n-1})$$

- ▶ Probabilities from a **language model**; processing effort at a word scales with its surprisal
- ▶ **Easy to compute** for arbitrary sentences
- ▶ Surprisal **predicts reading times well** on naturalistic text
(e.g., Demberg & Keller 2008; Smith & Levy 2013; Shain et al. 2024)

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(e.g., Demberg & Keller 2008; Smith & Levy 2013; Shain et al. 2024)
- ▶ But it **underpredicts syntactic disambiguation effects**
(van Schijndel & Linzen 2021; **SAP dataset** by Huang et al. 2024; Timkey et al. 2026)

Surprisal underpredicts disambiguation difficulty

(b) Results using log RTs

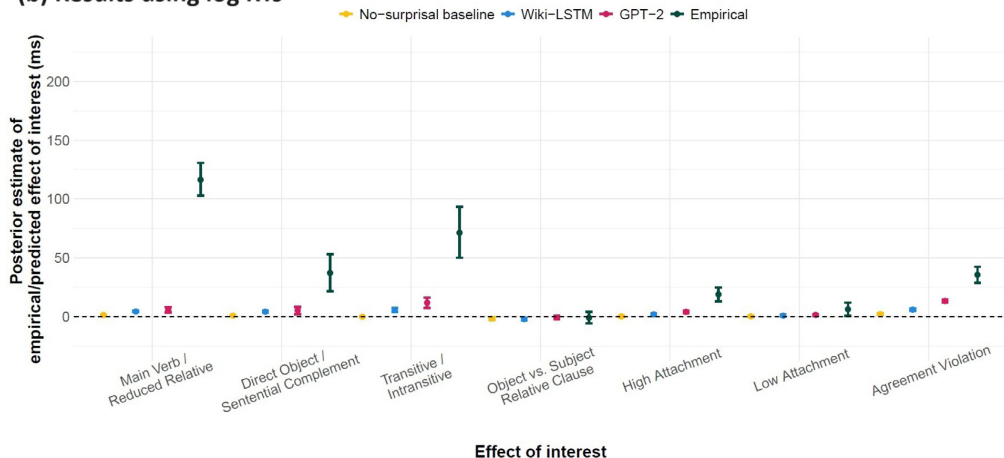


Fig. A.2 (b), Huang et al. (2024)

Can we compute **alternative**, theoretically motivated metrics that operate on the **same level as surprisal**?

Same level, i.e.:

- ▶ Word-by-word numerical predictions on arbitrary sentences
- ▶ **Automatically computable**: no hand-coding required
- ▶ Linking hypothesis: a simple scaling parameter to fit RTs

Predictive Metrics

1. **(Plain) surprisal** (GPT-2 small; the reference point)
2. **Resource-rational lossy context surprisal**
3. **Attention entropy**
4. **DLT integration cost** (+ reanalysis cost)
5. **ACT-R interference** (+ reanalysis cost)
6. **BERT interference** (+ dependency distance + reanalysis cost)

All of them are computable **hands-off** for arbitrary input, just like surprisal.

Predictive metrics: Resource-rational lossy context surprisal

- ▶ Surprisal, but computed on a **distorted memory representation** (Hahn et al. 2022; Futrell et al. 2020)
- ▶ Here: **20%, 50% and 80% deletion rates** (using pretrained models by McCurdy & Hahn 2024)

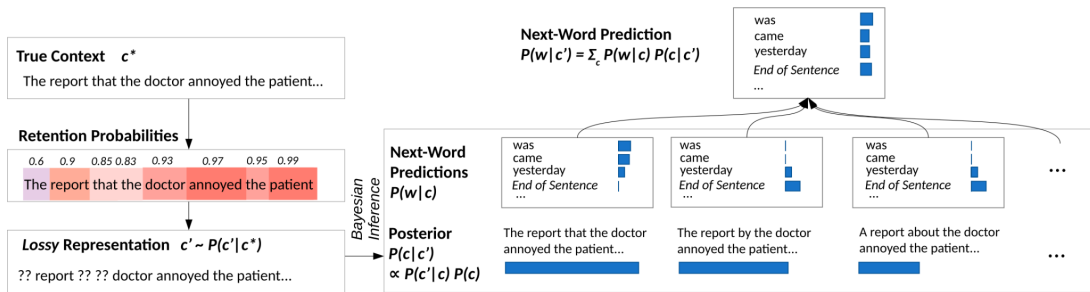


Fig. 2, Hahn et al. (2022)

Predictive metrics: DLT integration cost + reanalysis cost

Gibson (1998; 2000): Dependency Locality Theory

- ▶ Processing effort = **integration cost**: summed **dependency lengths** in terms of # discourse referents, from **Stanza** (Qi et al. 2020) dependency parses
- ▶ **Reanalysis cost** (auxiliary metric): how much does the parse so far **change** when w_n arrives, compared to the state at w_{n-1} ? (cf. Sturt, Pickering & Crocker 1999; Gorrell 1995)
+1 per word whose incoming or outgoing dependency arrow(s) just changed

Predictive metrics: Retrieval- and interference-based accounts

- ▶ **Attention entropy**: Transformer attention-head entropy from GPT-2 as a proxy for **cue-based retrieval** (Ryu & Lewis 2025; Lewis & Vasishth 2005), using their same 20 syntactically relevant heads
- ▶ **ACT-R interference**: simplified Lewis & Vasishth (2005) model re-implementation (in `pyactr`; Brasoveanu & Dotlačil 2020); retrieval cost at a word from all leftward codependents, with **interference** from feature-similar intervening words (`spaCy` + WordNet features; Honnibal et al. 2020; Miller 1995)
- ▶ **BERT interference**: same idea, but cosine similarity between **BERT embeddings** (Devlin et al. 2019) of a word and its left codependents, plus a separate dependency-distance predictor

ACT-R and BERT interference are complemented with reanalysis cost, just like DLT.

Data and Statistical Models

The SAP self-paced reading dataset (Huang et al. 2024)

N = 2,000 participants (post-exclusion)

Experimental design	Conditions
MV/RR garden paths	ambiguous unambiguous
NP/S garden paths	ambiguous unambiguous
NP/Z garden paths	ambiguous unambiguous
Relative clause asymmetry	subject object
Attachment ambiguity	high low ambiguous
Agreement violation	violation no-violation

Naturalistic fillers were also included, but we ignore these *for now*.

Statistical models & approach

Compare **log-normal** models on **spillover-region reading times**, one metric at a time:

Baseline + **one metric of interest** (+ auxiliary metrics)

```
RT ~ trialOrder + lengthCritical + lengthSpillover  
  + logFreqCritical + logFreqSpillover  
  + metricCritical + metricSpillover  
  + reanalysisCritical + reanalysisSpillover (DLT / ACT-R / BERT only)  
  + depDistanceCritical + depDistanceSpillover (BERT only)  
  + (1|subject) + (1|item) + (1|design)
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Coefficients of non-baseline metrics are reparameterized as $\beta = \gamma^2$ (using TMB): effect can only be ≥ 0 .

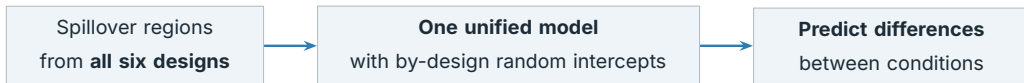
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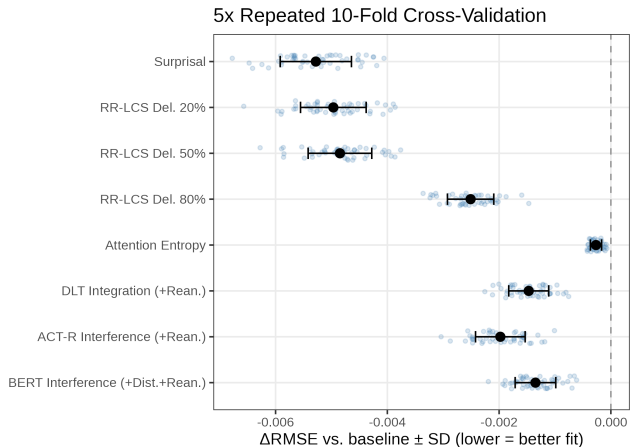
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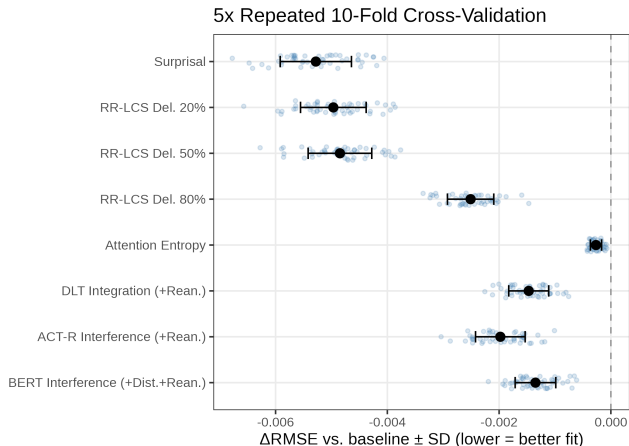


Results

No metric beats surprisal on held-out fit

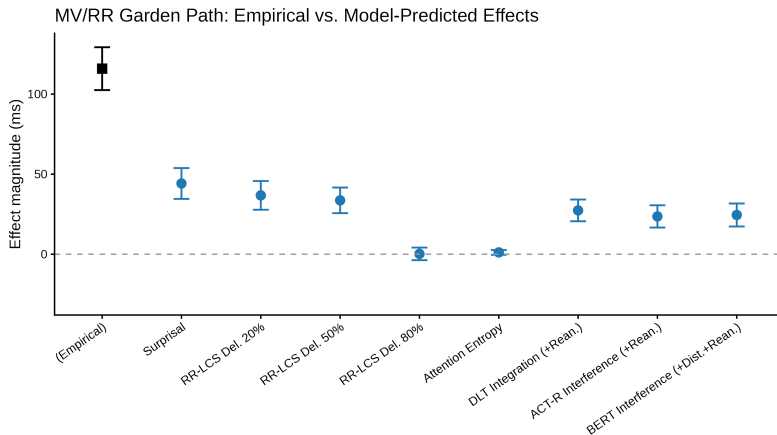


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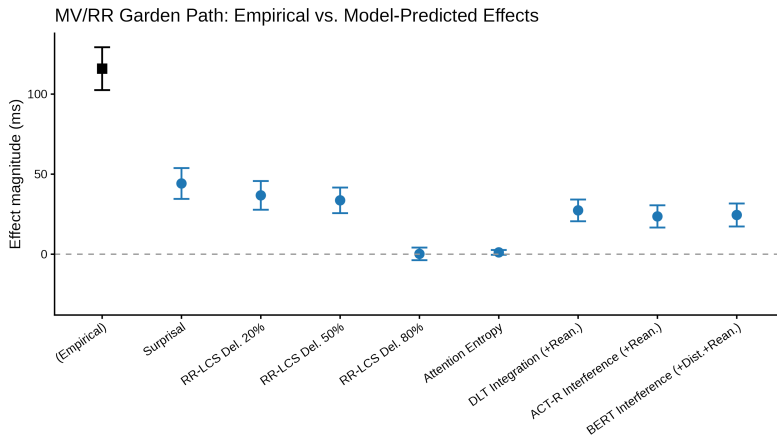


Surprisal's advantage over each alternative is **significant in every comparison**: paired t -tests with the Nadeau & Bengio (2003) correction for repeated k -fold CV, all $p < 0.003$ (closest competitor, RR-LCS Del. 20%: $t(49) = 3.2$, $p = 0.0024$)

MV/RR garden paths: empirical vs. model-predicted

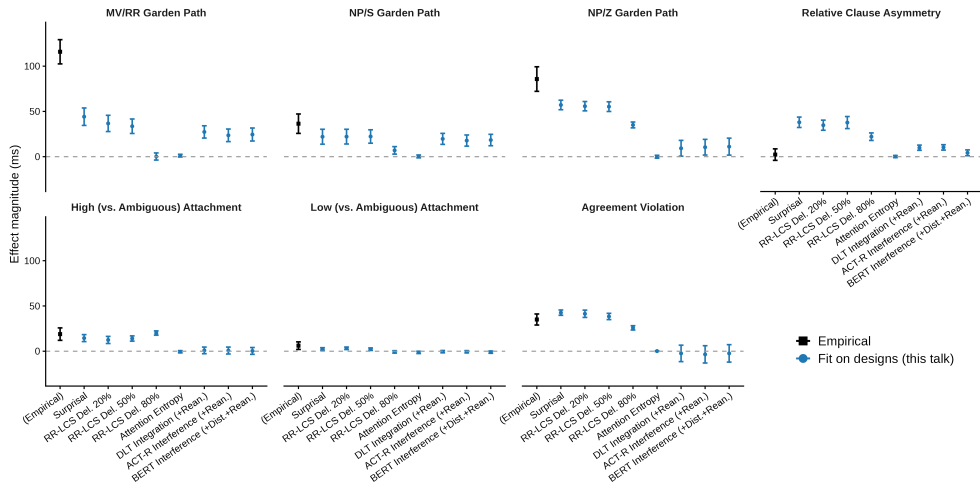


MV/RR garden paths: empirical vs. model-predicted



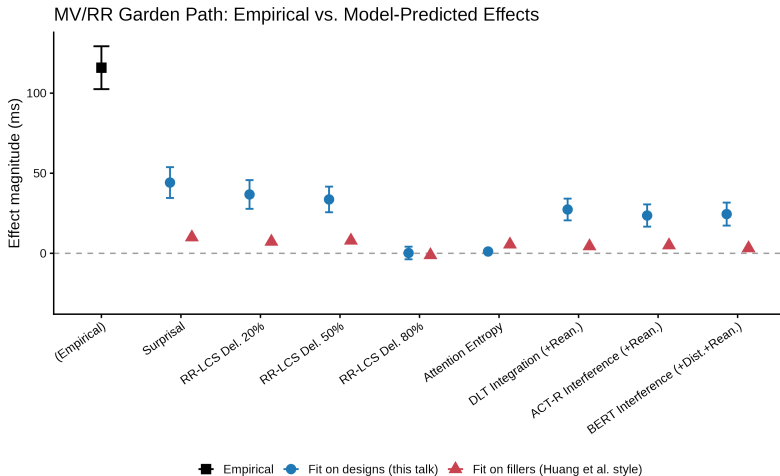
Underprediction is less dramatic than in Huang et al.'s (2024) analysis,
because **we fit on designs**, not fillers.

All seven contrasts



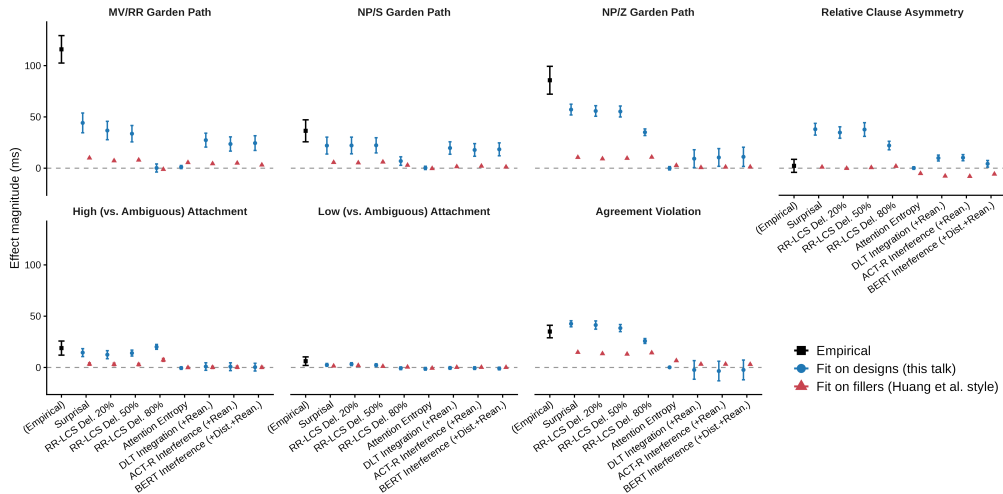
No metric systematically outperforms surprisal: others fare similarly or worse.

What if we had fit on fillers, like Huang et al.?



Filler-fit effects shrink toward zero for every metric.

What if we had fit on fillers, like Huang et al.?



No metric beats surprisal under **filler-fitting** either.

Conclusion

Discussion and conclusion

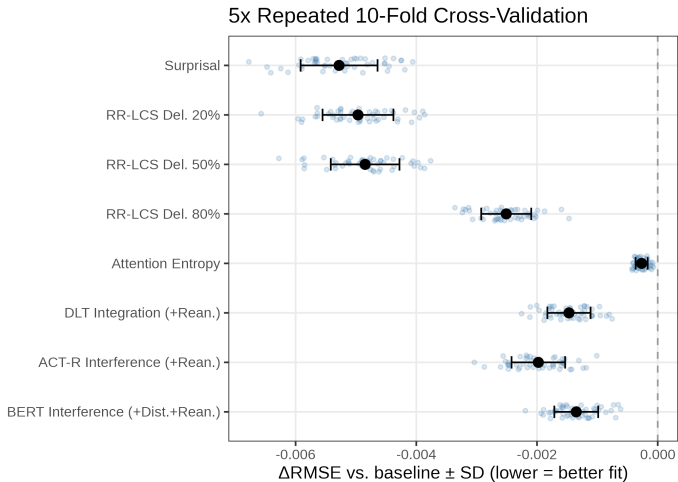
- ▶ **NLP tools** can operationalize predictive metrics of processing effort beyond surprisal, motivated by **memory-based**, **reanalysis-based** or hybrid **expectation-memory** accounts
- ▶ However, **no assessed metric outperformed surprisal** on the SAP dataset
- ▶ Among models given **no other information than the text itself**, surprisal still shows the relatively **highest predictive performance**

Discussion and conclusion

- ▶ **NLP tools** can operationalize predictive metrics of processing effort beyond surprisal, motivated by **memory-based**, **reanalysis-based** or hybrid **expectation-memory** accounts
 - ▶ However, **no assessed metric outperformed surprisal** on the SAP dataset
 - ▶ Among models given **no other information than the text itself**, surprisal still shows the relatively **highest predictive performance**
-
- ▶ We compared metrics of **one particular class**: automatically computable from text only
 - ▶ So we do **not show** that surprisal beats richer **process models** with more parameters and access to symbolic information beyond text (e.g., condition labels)
 - ▶ Paape, Linzen & Vasishth's (2026) latent-mixture model **outperforms** a mixture-free surprisal model under cross-validation, on the SAP dataset

I want to thank **Michael Hahn** and his research group
for helping me to compute RR-LCS predictions
for the sentences examined in this talk :)

Thank you for your attention!



Surprisal still sets the bar
**among automatically
computable metrics.**

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